NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

23 July 2026 | 09:46 PM

Session ID: 22 **Modules Used:** X-Ray Analysis **Date:** 23 July 2026 | 09:46 PM

X-Ray Analysis

Detected Region: Chest

Finding	Confidence
Pleural Effusion - Right side	High
Normal left lung	High
Normal cardiac silhouette	High
Normal mediastinum	High

■ **Urgency Level: Urgent — Same day attention required**

Recommended Tests

- Chest ultrasound or CT chest with contrast for better characterization of pleural effusion
- Thoracentesis with pleural fluid analysis (cell count, protein, LDH, glucose, cultures, cytology)
- Complete blood count and comprehensive metabolic panel
- BNP or NT-proBNP if cardiac etiology suspected
- Consider CT chest if malignancy is in differential diagnosis

Suggested Treatment

- Identify and treat underlying cause (cardiac failure, infection, malignancy, etc.)
- Monitor fluid balance and vital signs
- Consider diuretics if heart failure is the suspected cause
- Therapeutic thoracentesis if effusion is causing respiratory symptoms or hemodynamic compromise
- Antibiotics if infectious etiology is identified

Next Steps

- Perform clinical examination to correlate with imaging findings and assess for respiratory distress
- Arrange pleural ultrasound to assess effusion characteristics and guide thoracentesis if indicated
- Send pleural fluid for complete analysis including culture and sensitivity
- Obtain relevant history regarding dyspnea, fever, chest pain, and past medical history
- Consider referral to respiratory specialist or pulmonology if diagnosis remains unclear

Summary: *Chest X-ray demonstrates a right-sided pleural effusion of moderate size with blunting of the right costophrenic angle. The left lung, cardiac silhouette, and mediastinum are within normal limits.*

■ ***DISCLAIMER:*** *This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.*